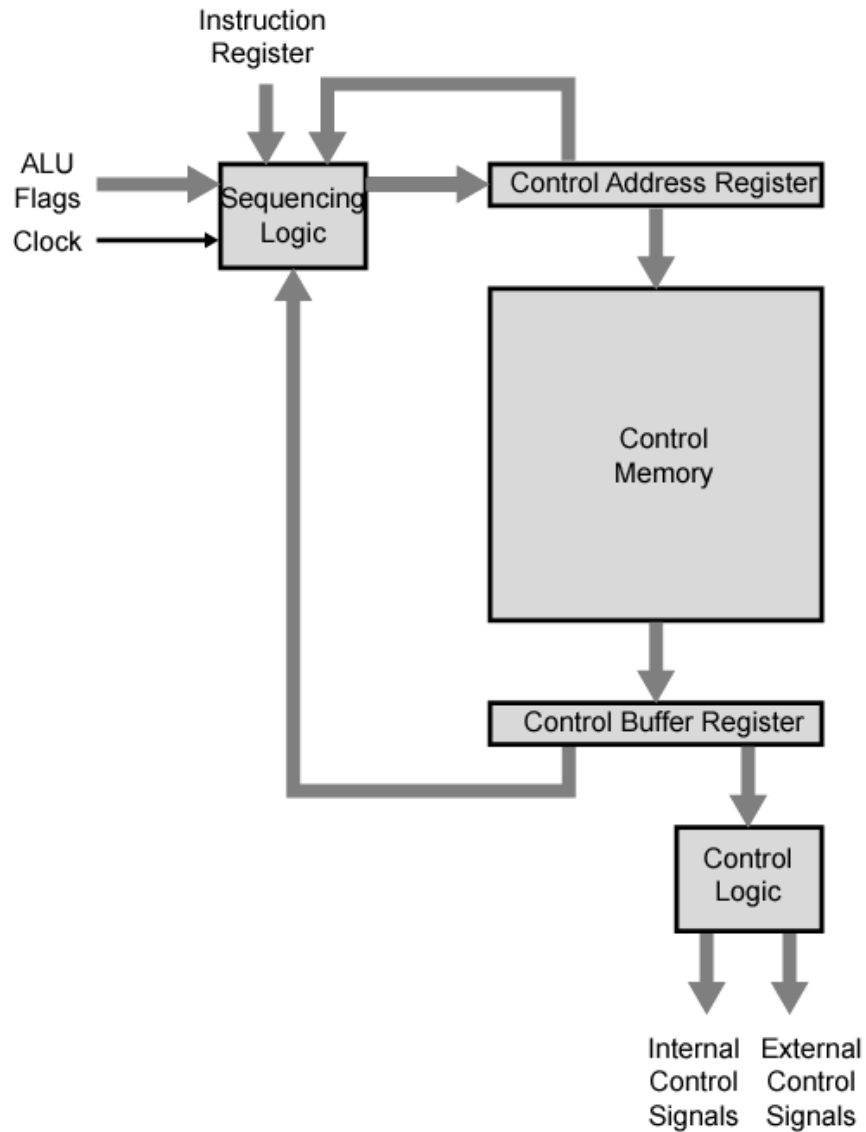


William Stallings
Computer Organization
and Architecture
7th Edition

Chapter 17
Micro-programmed Control

Control Unit Organization



Micro-programmed Control

- Use sequences of instructions (see earlier notes) to control complex operations
- Called micro-programming or firmware

Implementation (1)

- All the control unit does is generate a set of control signals
- Each control signal is on or off
- Represent each control signal by a bit
- Have a control word for each micro-operation
- Have a sequence of control words for each machine code instruction
- Add an address to specify the next micro-instruction, depending on conditions

Implementation (2)

- Today's large microprocessor
 - Many instructions and associated register-level hardware
 - Many control points to be manipulated
- This results in control memory that
 - Contains a large number of words
 - co-responding to the number of instructions to be executed
 - Has a wide word width
 - Due to the large number of control points to be manipulated

Micro-program Word Length

- Based on 3 factors
 - Maximum number of simultaneous micro-operations supported
 - The way control information is represented or encoded
 - The way in which the next micro-instruction address is specified

Micro-instruction Types

- Each micro-instruction specifies single (or few) micro-operations to be performed
 - (*vertical* micro-programming)
- Each micro-instruction specifies many different micro-operations to be performed in parallel
 - (*horizontal* micro-programming)

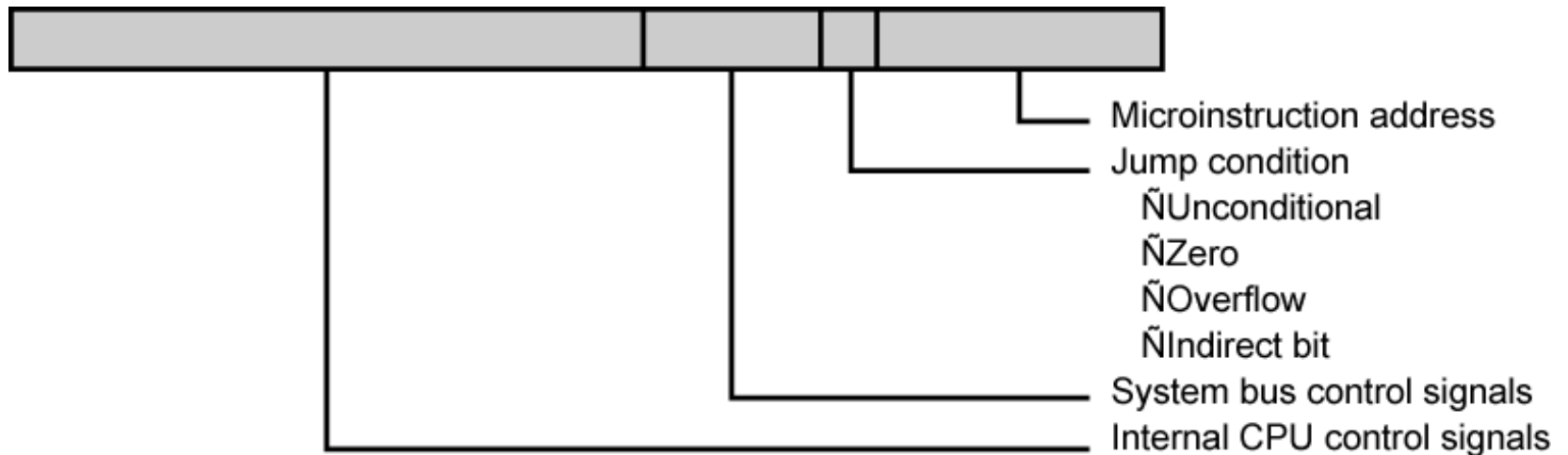
Vertical Micro-programming

- Width is narrow
- n control signals encoded into $\log_2 n$ bits
- Limited ability to express parallelism
- Considerable encoding of control information requires external memory word decoder to identify the exact control line being manipulated

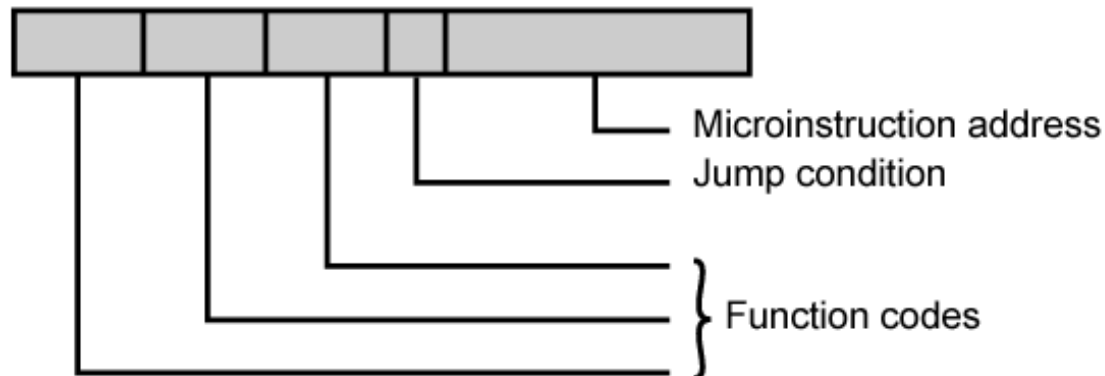
Horizontal Micro-programming

- Wide memory word
- High degree of parallel operations possible
- Little encoding of control information

Typical Microinstruction Formats



(a) Horizontal microinstruction

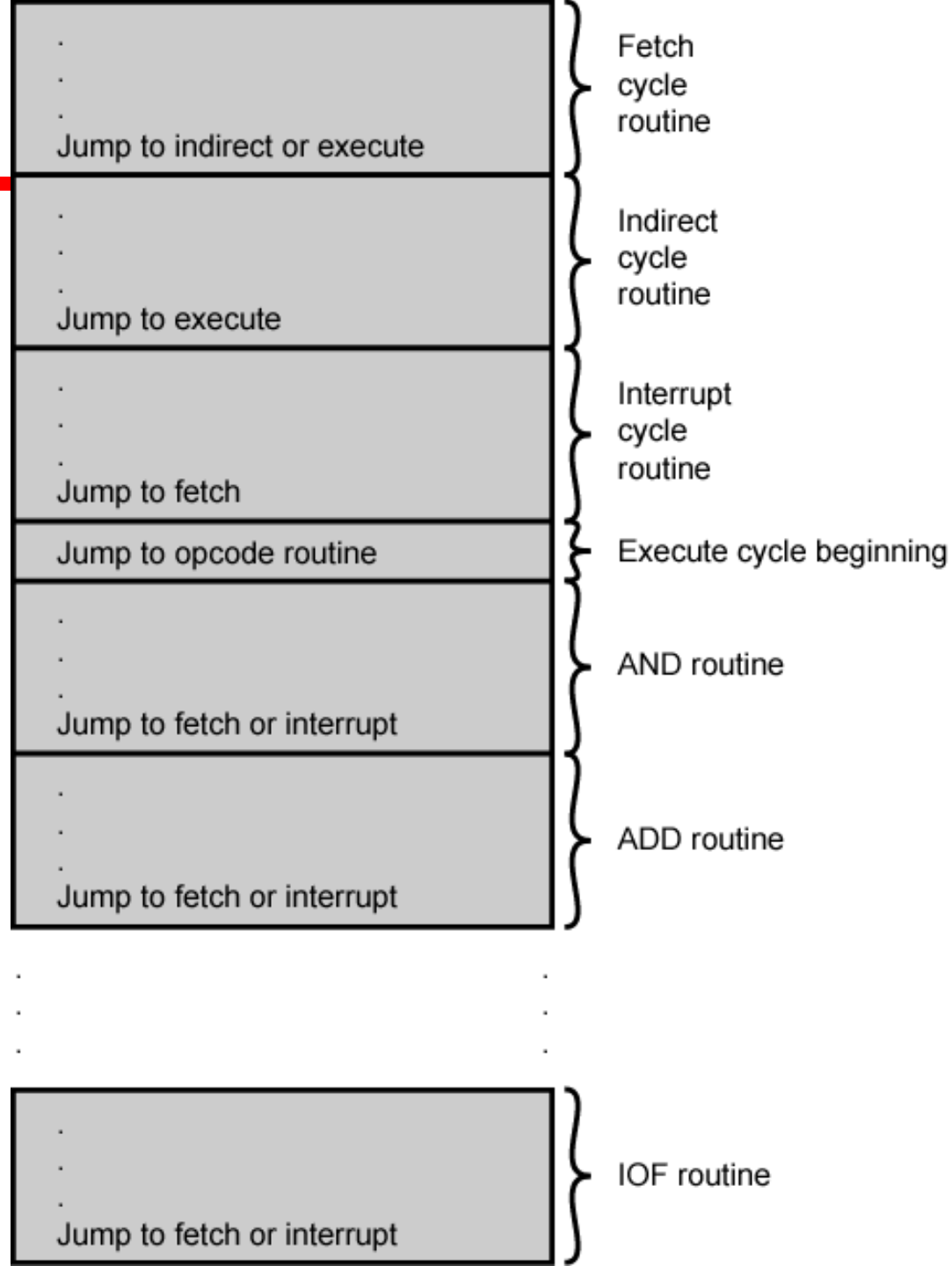


(b) Vertical microinstruction

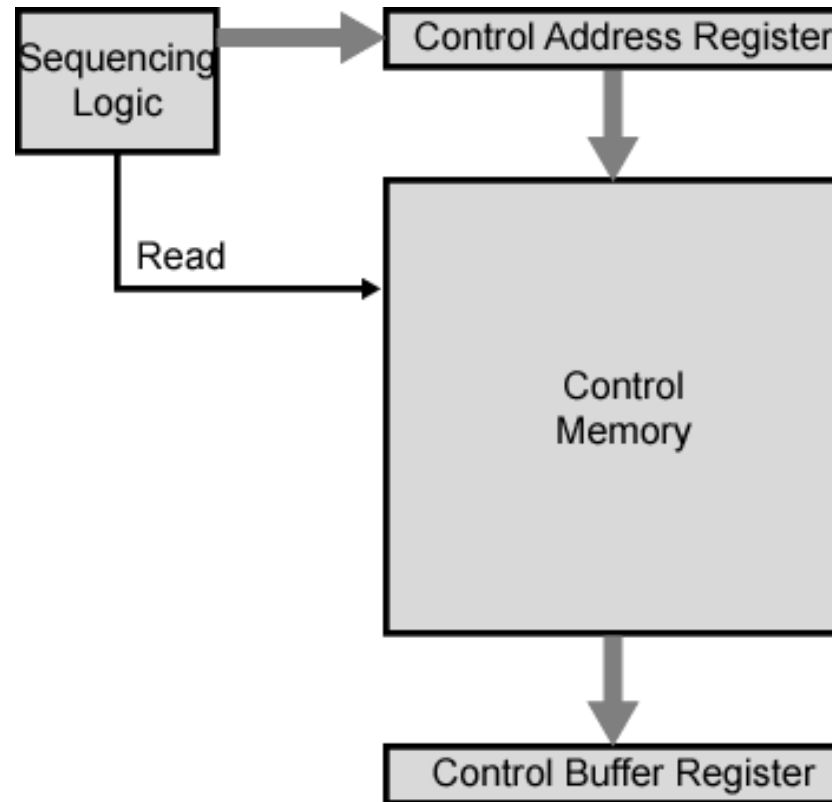
Compromise

- Divide control signals into disjoint groups
- Implement each group as separate field in memory word
- Supports reasonable levels of parallelism without too much complexity

Organization of Control Memory



Control Unit



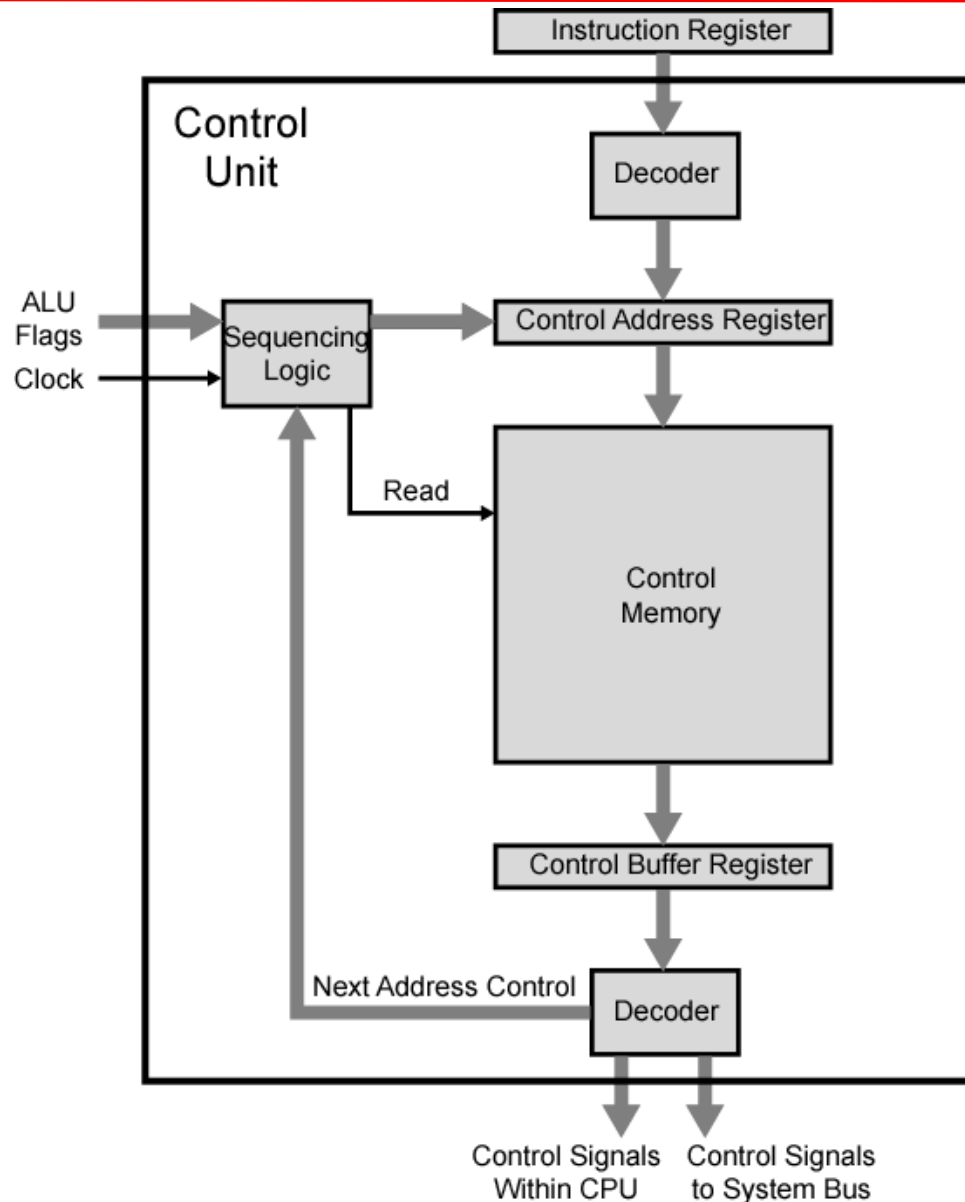
Control Unit Function

- Sequence login unit issues read command
- Word specified in control address register is read into control buffer register
- Control buffer register contents generates control signals and next address information
- Sequence login loads new address into control buffer register based on next address information from control buffer register and ALU flags

Next Address Decision

- Depending on ALU flags and control buffer register
 - Get next instruction
 - Add 1 to control address register
 - Jump to new routine based on jump microinstruction
 - Load address field of control buffer register into control address register
 - Jump to machine instruction routine
 - Load control address register based on opcode in IR

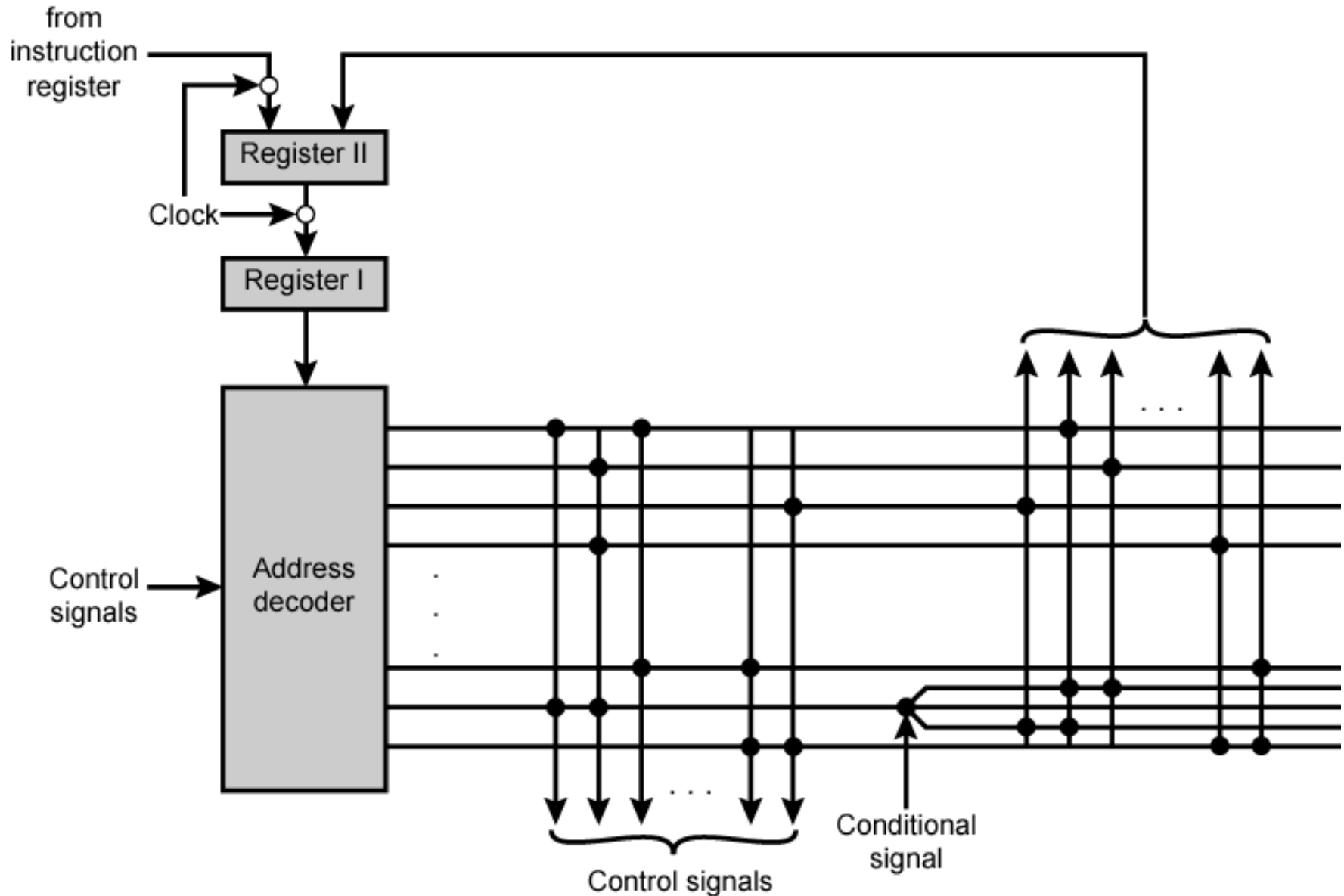
Functioning of Microprogrammed Control Unit



Wilkes Control

- 1951
- Matrix partially filled with diodes
- During cycle, one row activated
 - Generates signals where diode present
 - First part of row generates control
 - Second generates address for next cycle

Wilkes's Microprogrammed Control Unit



Advantages and Disadvantages of Microprogramming

- Simplifies design of control unit
 - Cheaper
 - Less error-prone
- Slower

Tasks Done By Microprogrammed Control Unit

- Microinstruction sequencing
- Microinstruction execution
- Must consider both together

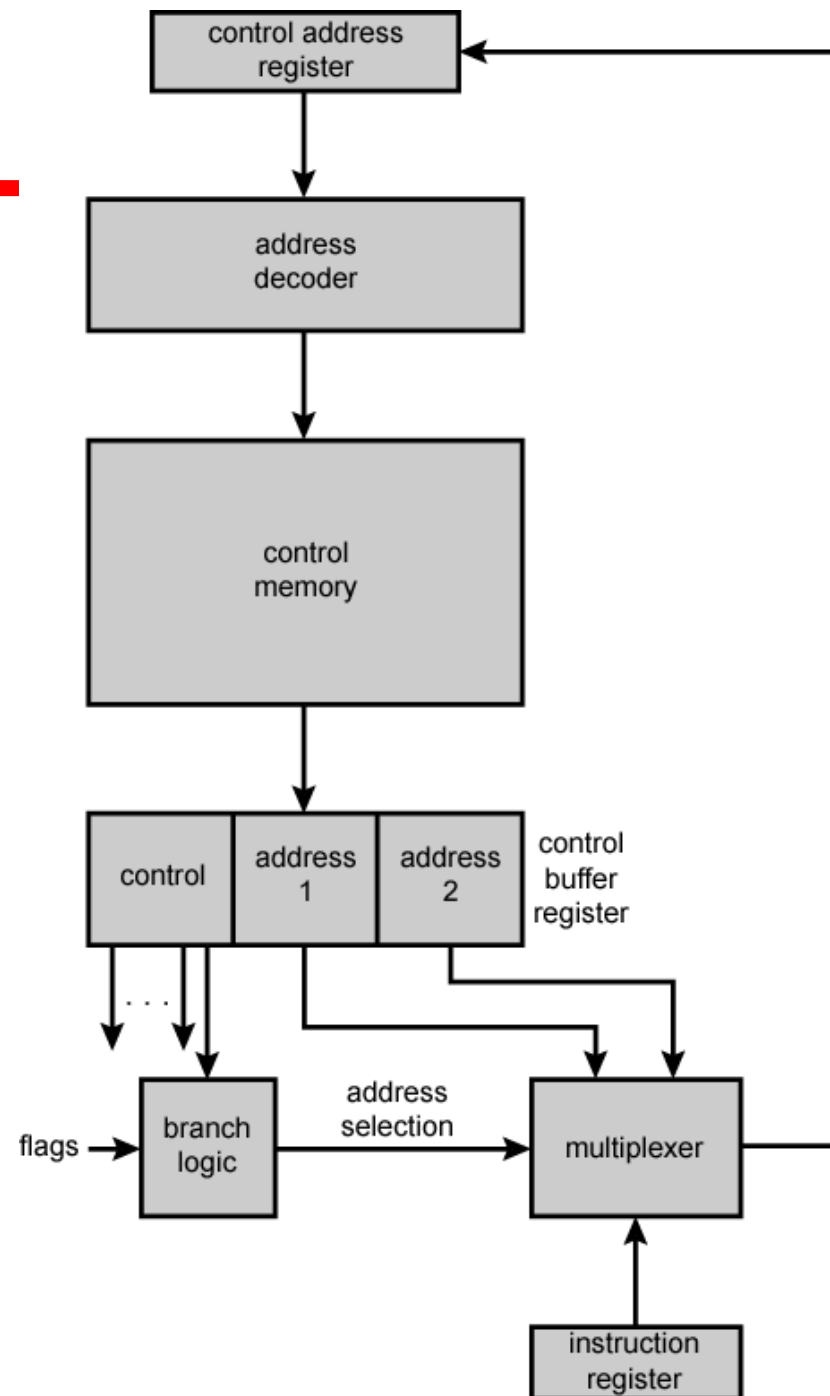
Design Considerations

- Size of microinstructions
- Address generation time
 - Determined by instruction register
 - Once per cycle, after instruction is fetched
 - Next sequential address
 - Common in most designed
 - Branches
 - Both conditional and unconditional

Sequencing Techniques

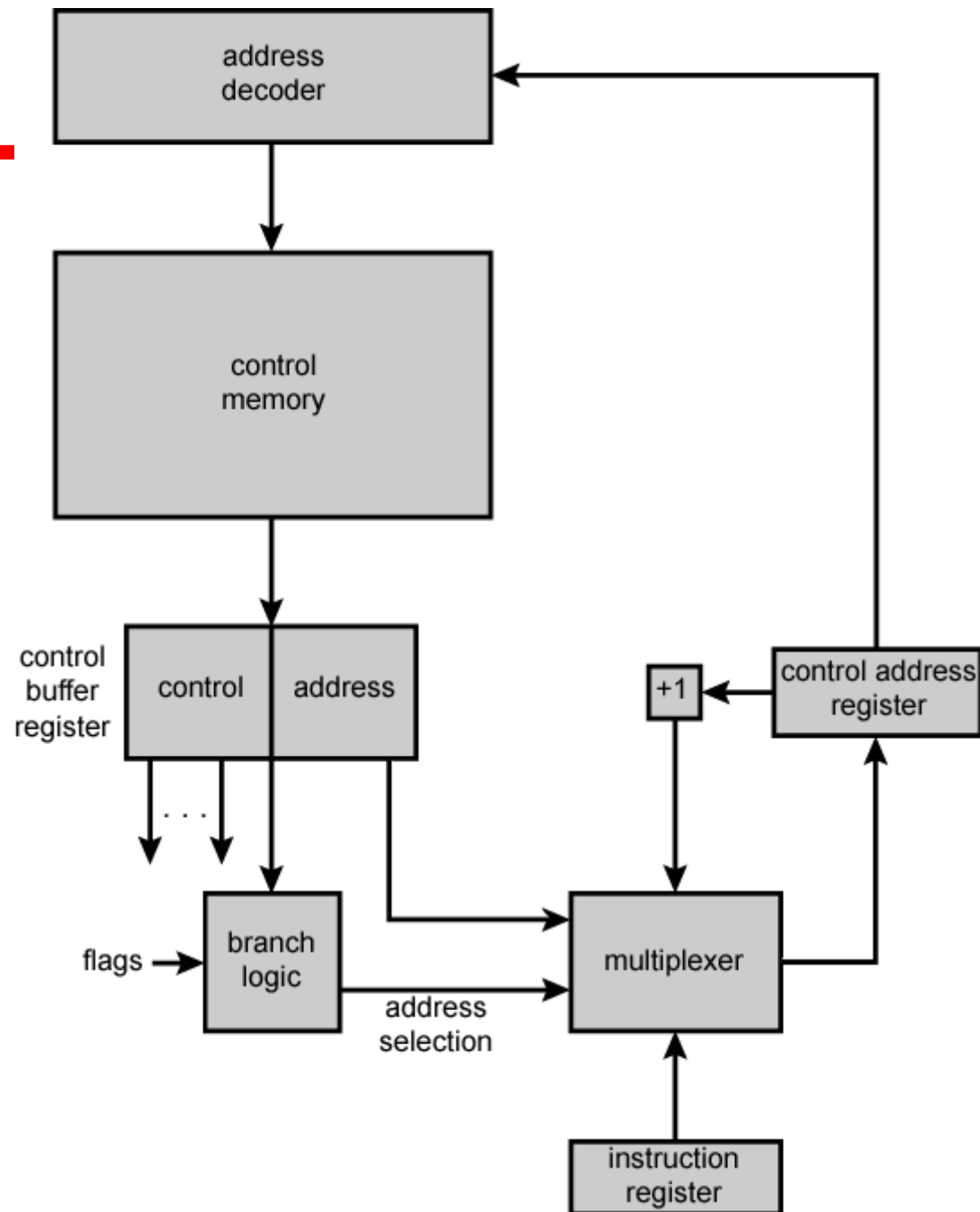
- Based on current microinstruction, condition flags, contents of IR, control memory address must be generated
- Based on format of address information
 - Two address fields
 - Single address field
 - Variable format

Branch Control Logic: Two Address Fields

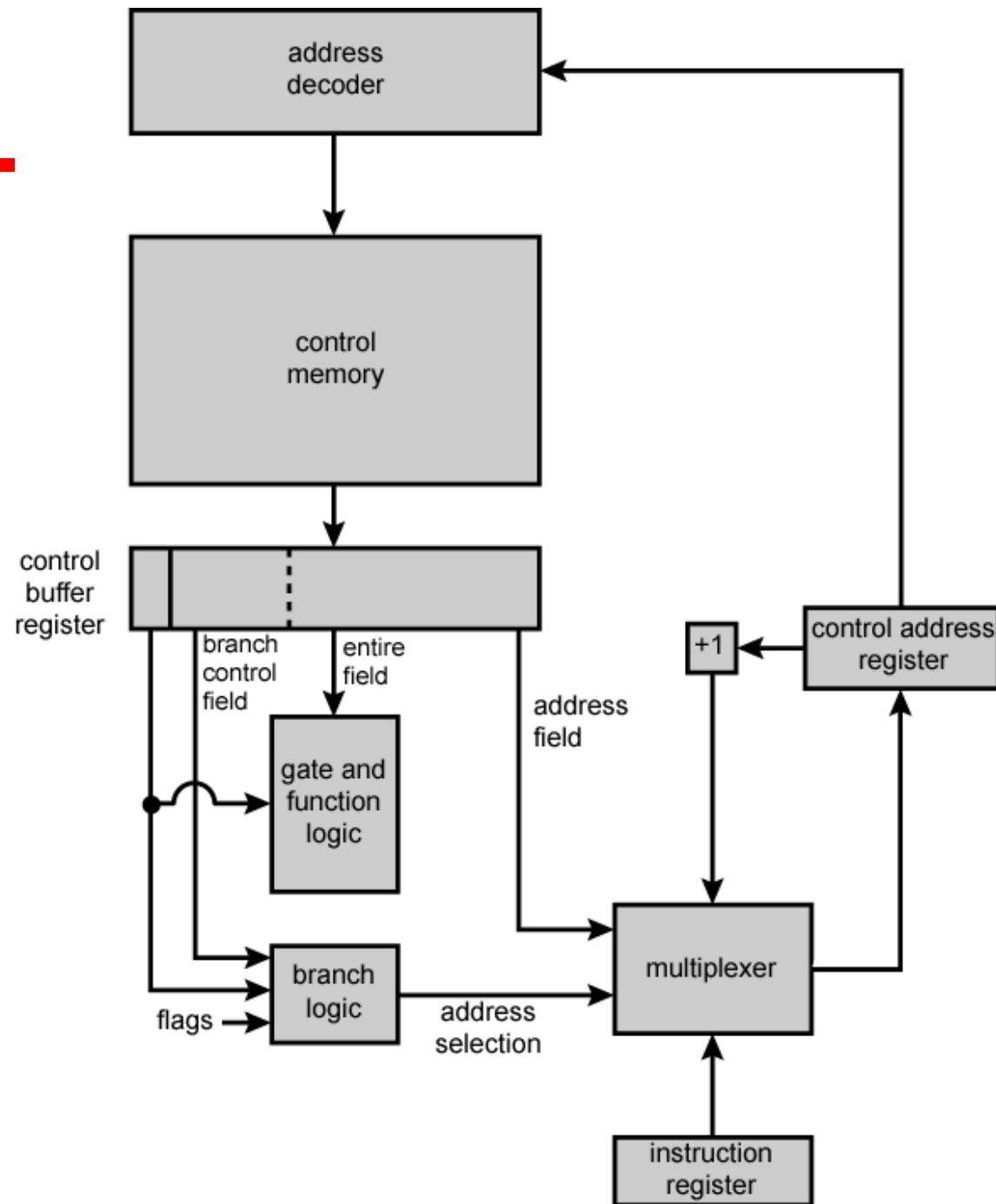


Branch Control

Logic: Single Address Field



Branch Control Logic: Variable Format



Address Generation

Explicit

Implicit

Two-field

Mapping

Unconditional Branch

Addition

Conditional branch

Residual control

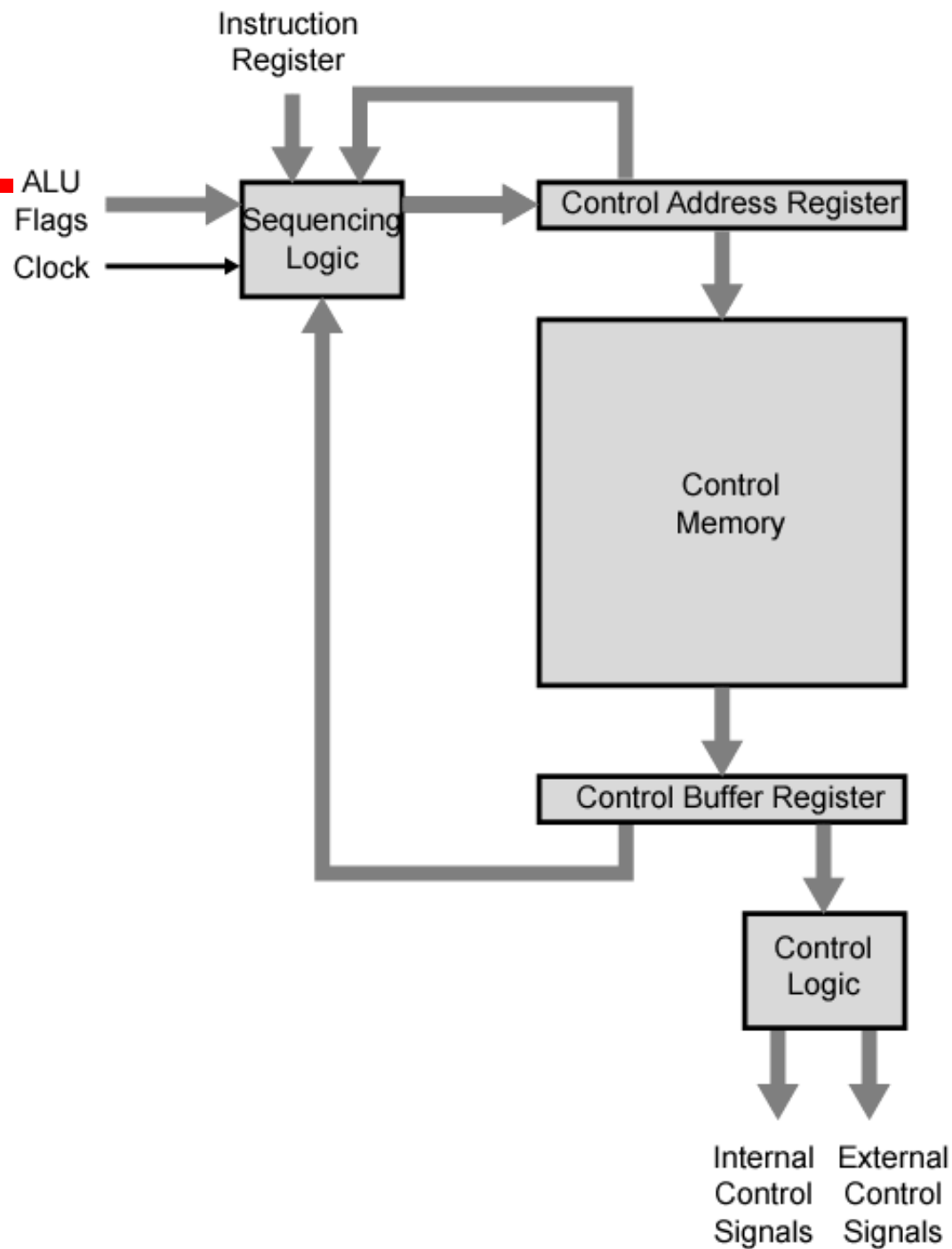
Execution

- The cycle is the basic event
- Each cycle is made up of two events
 - Fetch
 - Determined by generation of microinstruction address
 - Execute

Execute

- Effect is to generate control signals
- Some control points internal to processor
- Rest go to external control bus or other interface

Control Unit Organization



A Taxonomy of Microinstructions

- Vertical/horizontal
- Packed/unpacked
- Hard/soft microprogramming
- Direct/indirect encoding

Improvements over Wilkes

- Wilkes had each bit directly produced a control signal or directly produced one bit of next address
- More complex address sequencing schemes,
- using fewer microinstruction bits, are possible
- Require more complex sequencing logic module
- Control word bits can be saved by encoding and subsequently decoding control information

How to Encode

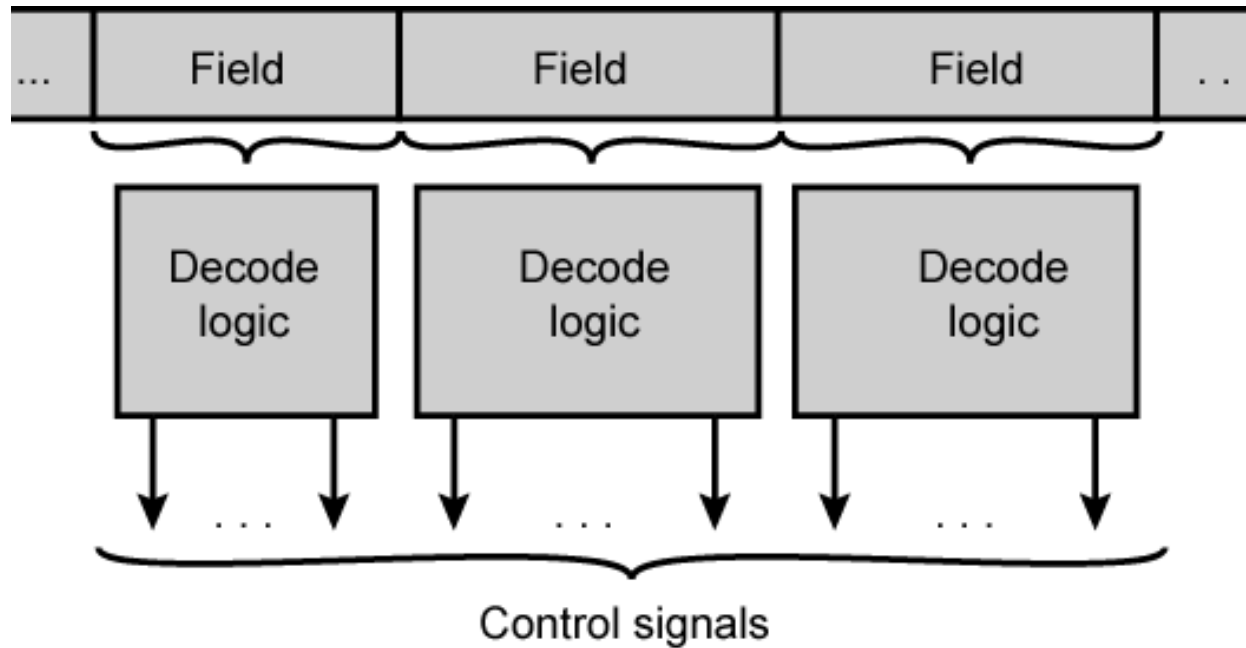
- K different internal and external control signals
- Wilkes's:
 - K bits dedicated
 - 2K control signals during any instruction cycle
- Not all used
 - Two sources cannot be gated to same destination
 - Register cannot be source and destination
 - Only one pattern presented to ALU at a time
 - Only one pattern presented to external control bus at a time
- Require $Q < 2K$ which can be encoded with $\log_2 Q < K$ bits
- Not done
 - As difficult to program as pure decoded (Wilkes) scheme
 - Requires complex slow control logic module
- Compromises
 - More bits than necessary used
 - Some combinations that are physically allowable are not possible to encode

Specific Encoding Techniques

- Microinstruction organized as set of fields
- Each field contains code
- Activates one or more control signals
- Organize format into independent fields
 - Field depicts set of actions (pattern of control signals)
 - Actions from different fields can occur simultaneously
- Alternative actions that can be specified by a field are mutually exclusive
 - Only one action specified for field could occur at a time

Microinstruction Encoding

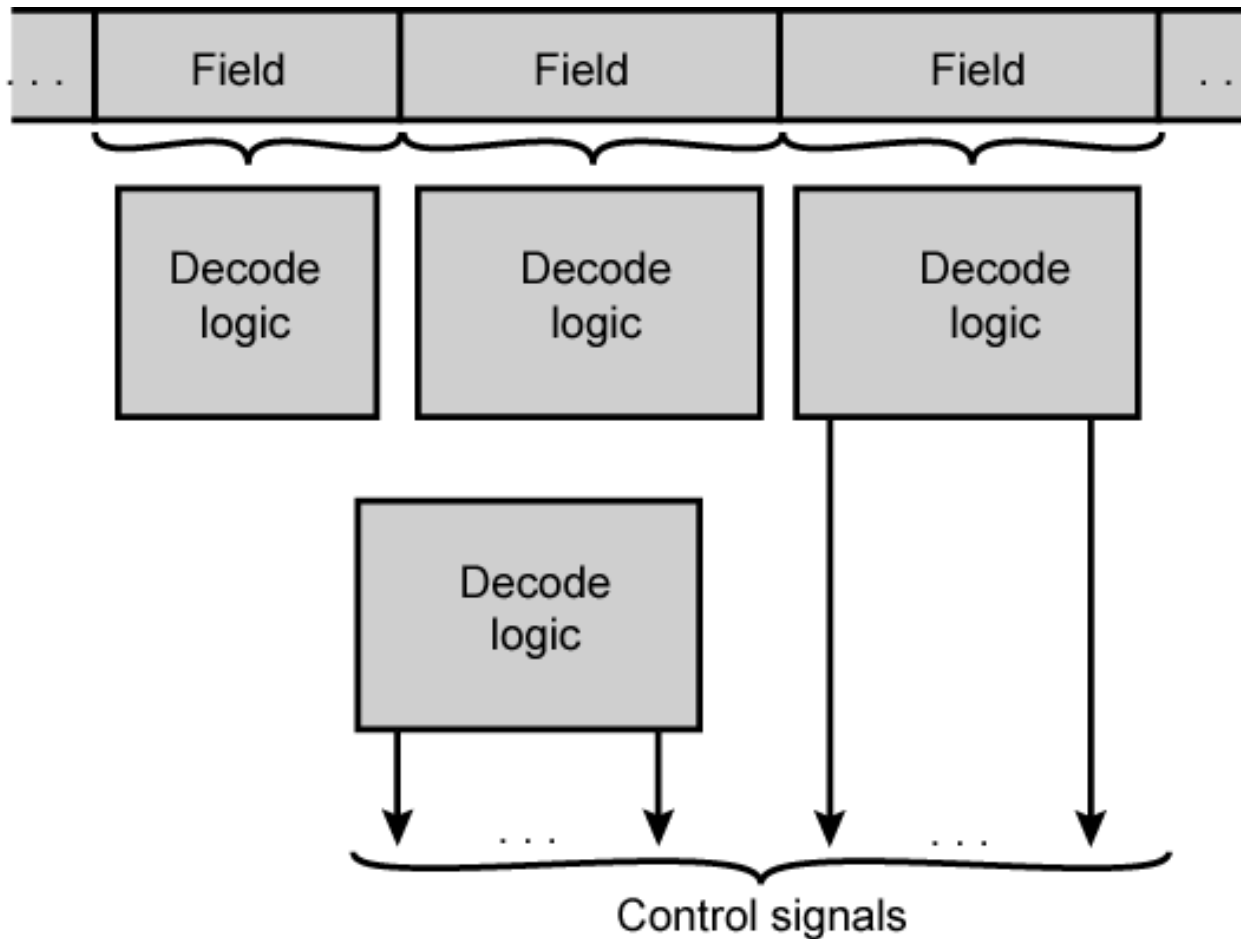
Direct Encoding



(a) Direct encoding

Microinstruction Encoding

Indirect Encoding



(b) Indirect encoding

Required Reading

- Stallings chapter 17